

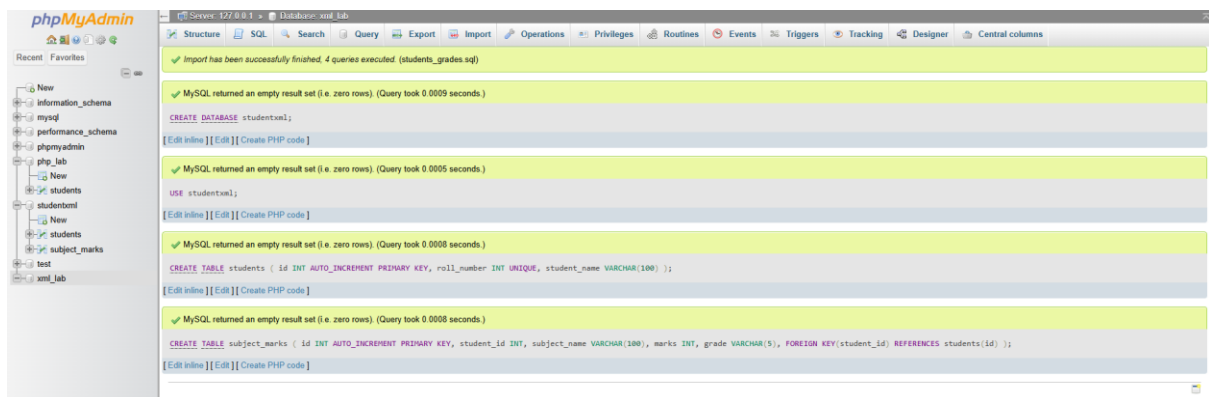
20CYS403 – Web Application Security

XML LAB

Ponvedica M S

CB.SC.U4CYS23034

1. Created 2 table – students and subject_marks under the database named studentxml



2. UI for the validation and calculating the marks

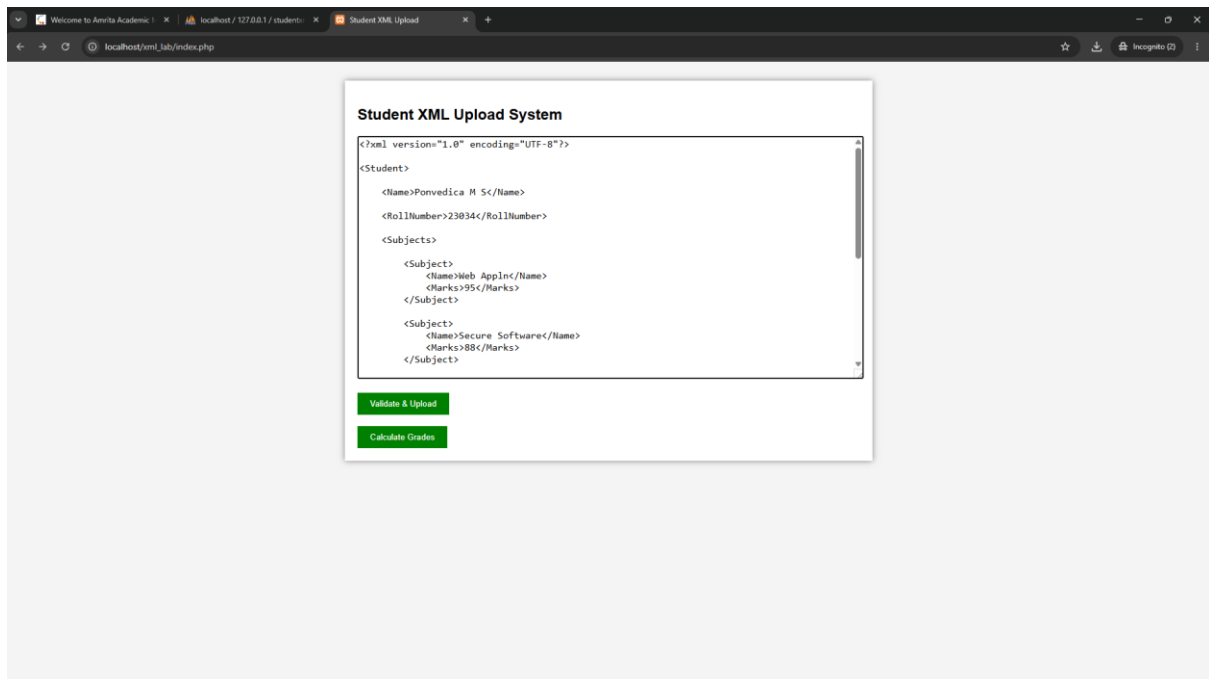
Student XML Upload System

Paste XML Here

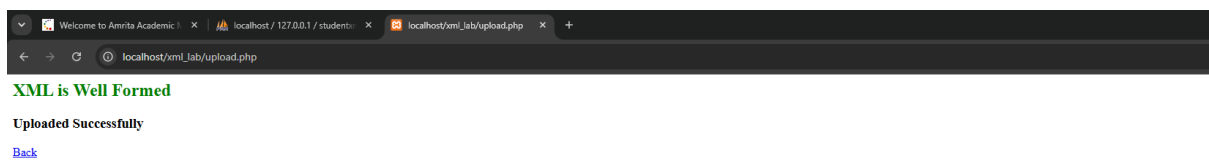
Validate & Upload

Calculate Grades

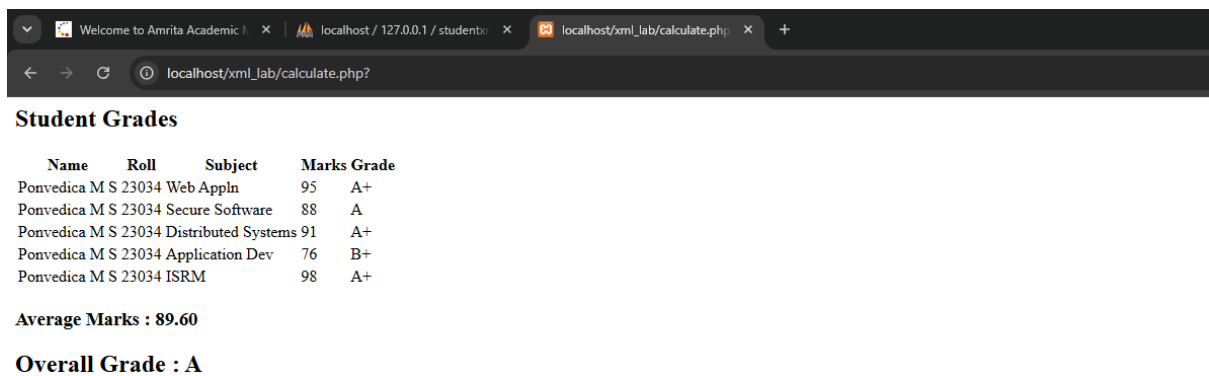
3. Now, pasting the correct validation of xml file and then validating



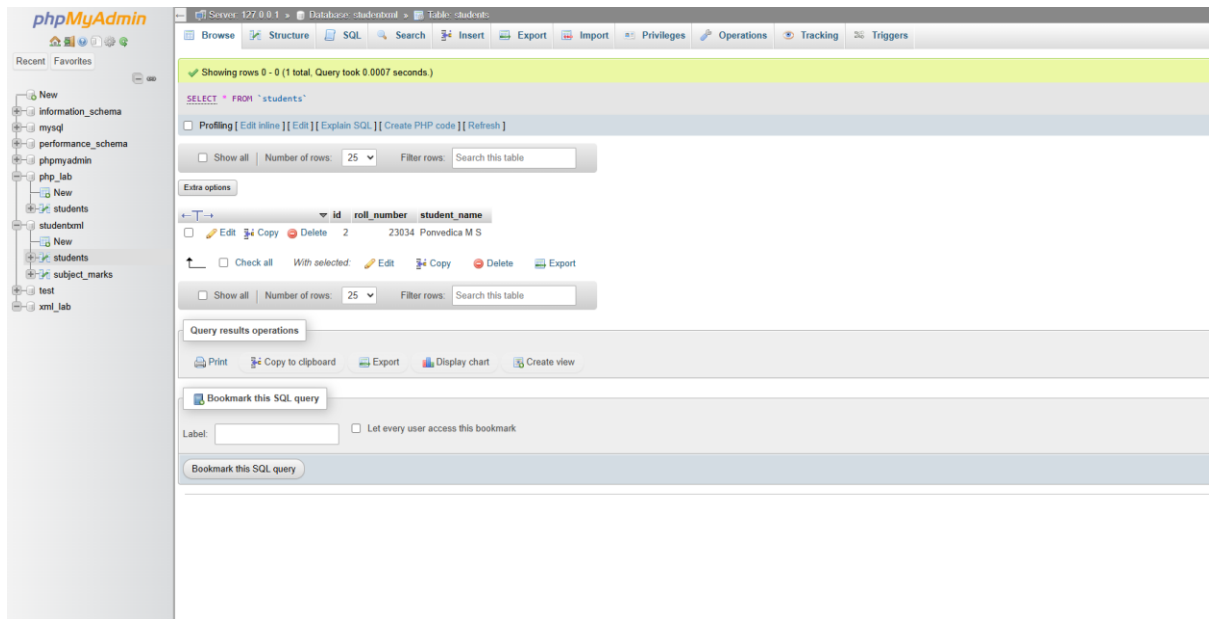
4. Since, it is well formed it is validated and the data are uploaded to the database successfully



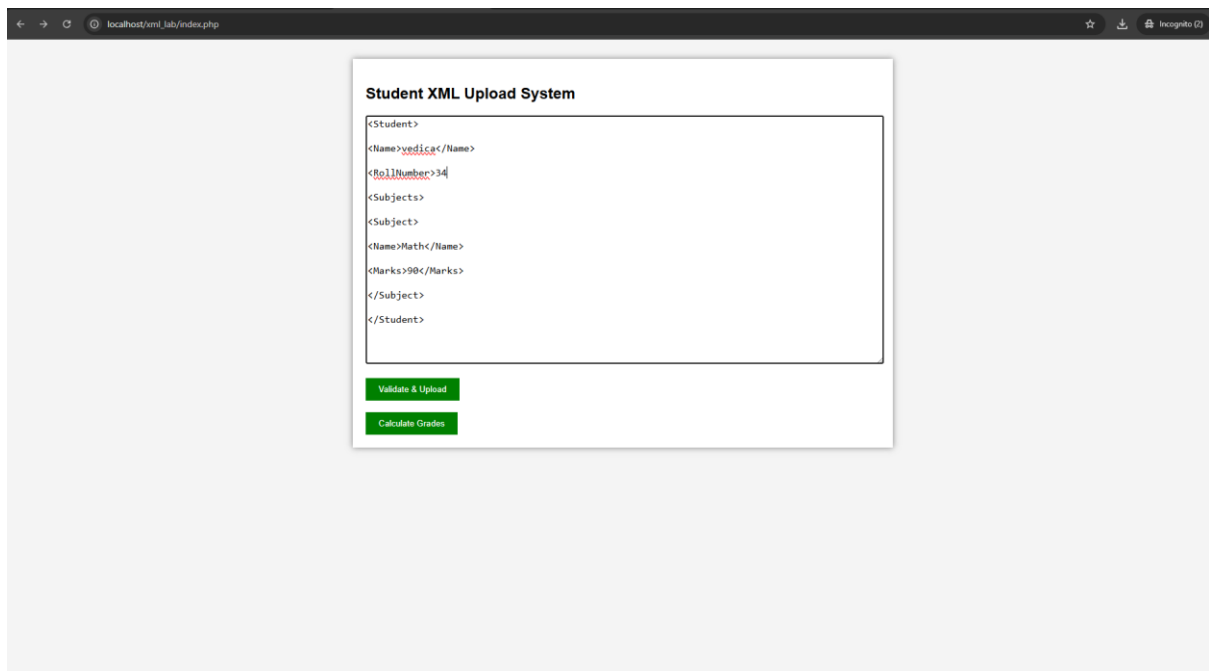
5. Now, if we click the calculate grades, it calculates and display it.



6. We can see the reflected data in the database



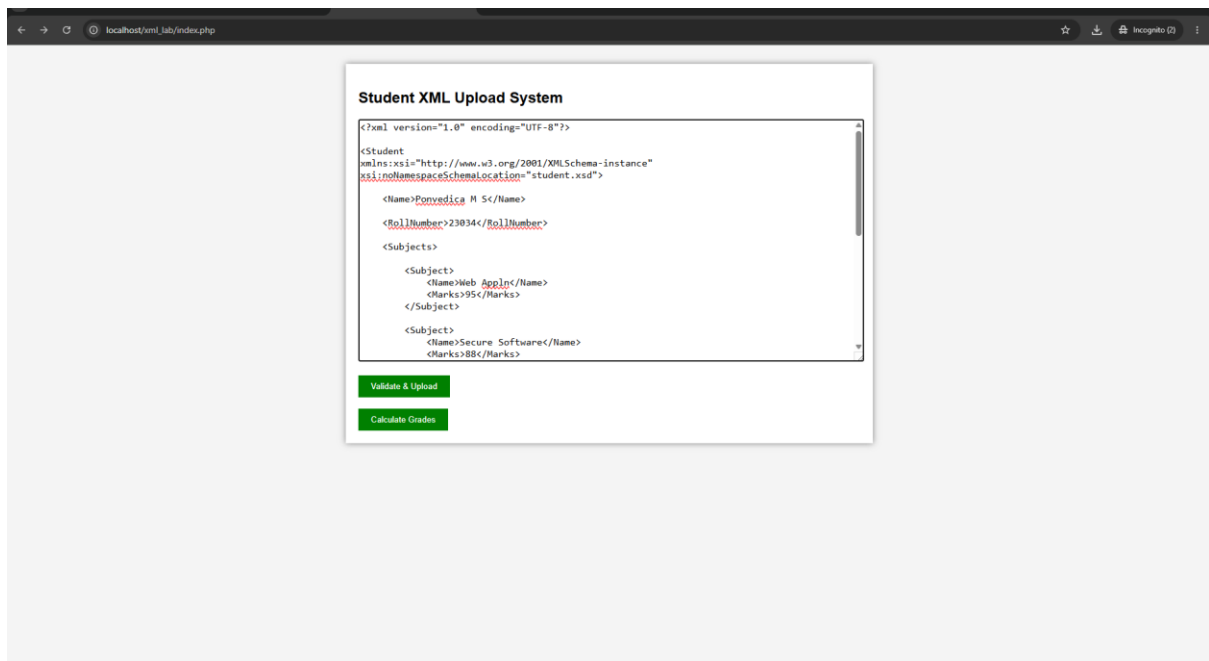
7. Not well formed xml is pasted to check for validation

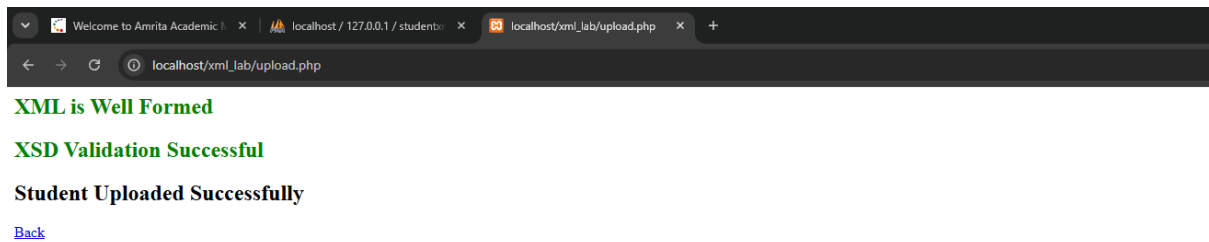


8. It displays the Error and mentions what error it is

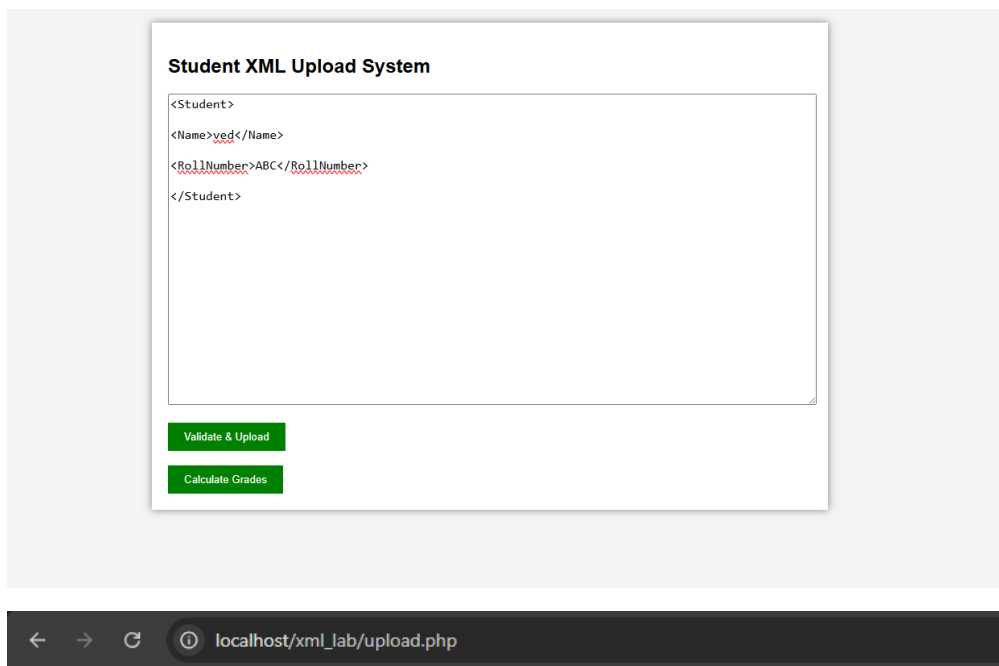


9. XSD validation also included now. – I added new .xsd file and then changed some code with respect to upload.php





10. Not well formed xsd but validated xml



XML is Well Formed

XML does NOT follow XSD

Element 'RollNumber': 'ABC' is not a valid value of the atomic type 'xs:integer'.
Element 'Student': Missing child element(s). Expected is (Subjects).

Student XML Upload System

```
<Student>
<Name>ved</Name>
<RollNumber>123</RollNumber>
</Student>
```

Validate & Upload

Calculate Grades

localhost/xml_lab/upload.php

XML is Well Formed

XML does NOT follow XSD

Element 'Student': Missing child element(s). Expected is (Subjects).

Codes :

SQL queries

```
CREATE DATABASE studentxml;

USE studentxml;

CREATE TABLE students (
    id INT AUTO_INCREMENT PRIMARY KEY,
    roll_number INT UNIQUE,
    student_name VARCHAR(100)
);

CREATE TABLE subject_marks (
    id INT AUTO_INCREMENT PRIMARY KEY,
```

```
student_id INT,  
subject_name VARCHAR(100),  
marks INT,  
grade VARCHAR(5),  
FOREIGN KEY(student_id) REFERENCES students(id)  
);
```

Calculate.php

```
<?php  
  
include("db.php");  
  
$result=$conn->query("SELECT students.student_name,  
students.roll_number,  
subject_marks.*  
  
FROM students  
  
JOIN subject_marks  
  
ON students.id=subject_marks.student_id");  
  
echo "<h2>Student Grades</h2>";  
  
echo "<table>";  
  
echo "<tr>  
  
<th>Name</th>  
  
<th>Roll</th>  
  
<th>Subject</th>  
  
<th>Marks</th>  
  
<th>Grade</th>  
  
</tr>";  
  
$total=0;  
  
$count=0;  
  
while($row=$result->fetch_assoc())  
{
```

```
$marks=$row['marks'];

if($marks>=90)
$grade="A+";

else if($marks>=80)
$grade="A";

else if($marks>=70)
$grade="B+";

else if($marks>=60)
$grade="B";

else if($marks>=50)
$grade="C";

else if($marks>=40)
$grade="D";

else
$grade="F";

$conn->query("UPDATE subject_marks
SET grade='$grade'
WHERE id=".$row['id']);

echo "<tr>";

echo "<td>".$row['student_name']. "</td>";

echo "<td>".$row['roll_number']. "</td>";

echo "<td>".$row['subject_name']. "</td>";

echo "<td>".$marks. "</td>";

echo "<td>".$grade. "</td>";

echo "</tr>";

$total+=$marks;

$count++;}

echo "</table>";

$avg=$total/$count;
```



```

echo "<h3>Average Marks : ".number_format($avg,2)."</h3>";

if($avg>=90)
$overall="A+";
else if($avg>=80)
$overall="A";
else if($avg>=70)
$overall="B+";
else if($avg>=60)
$overall="B";
else if($avg>=50)
$overall="C";
else if($avg>=40)
$overall="D";
else
$overall="F";

echo "<h2>Overall Grade : $overall</h2>";

?>

```

Db.php

```

<?php

$host="localhost";
$user="root";
$password="";
$dbname="studentxml";
$conn = new mysqli($host,$user,$password,$dbname);

if($conn->connect_error)
{ die("Connection Failed : ".$conn->connect_error);}

?>

```

Index.php

```

<!DOCTYPE html>
<html>

<head>

<title>Student XML Upload</title>

<link rel="stylesheet" href="style.css">

```

```
</head>

<body>

<div class="container">

<h2>Student XML Upload System</h2>

<form action="upload.php" method="post">

<textarea name="xml" rows="20" cols="80"
placeholder="Paste XML Here"></textarea>

<br><br>

<input type="submit" value="Validate & Upload">

</form>

<br>

<form action="calculate.php">

<input type="submit" value="Calculate Grades">

</form>

</div>

</body>

</html>
```

Invalid xml

```
<Student>

<Name>Vedica</Name>

<RollNumber>34

<Subjects>

<Subject>

<Name>Math</Name>

<Marks>90</Marks>
```

```
</Subject>
```

```
</Student>
```

Valid xml

```
<?xml version="1.0" encoding="UTF-8"?>

<Student
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:noNamespaceSchemaLocation="student.xsd">

    <Name>Ponvedica M S</Name>

    <RollNumber>23034</RollNumber>

    <Subjects>

        <Subject>
            <Name>Web Appln</Name>
            <Marks>95</Marks>
        </Subject>

        <Subject>
            <Name>Secure Software</Name>
            <Marks>88</Marks>
        </Subject>

        <Subject>
            <Name>Distributed Systems</Name>
            <Marks>91</Marks>
        </Subject>

        <Subject>
            <Name>Application Dev</Name>
            <Marks>76</Marks>
        </Subject>

        <Subject>
            <Name>ISRM</Name>
            <Marks>98</Marks>
        </Subject>

    </Subjects>

</Student>
```

CSS

```
body{

font-family:Arial;

background:#f4f4f4;

}

.container{

width:800px;

margin:auto;

background:white;

padding:20px;

margin-top:30px;

box-shadow:0px 0px 10px gray;

}

textarea{

width:100%;

font-size:16px;

}

input[type=submit]{

padding:10px 20px;

background:green;

color:white;

border:none;

cursor:pointer;

}
```

```
table{

width:100%;

border-collapse:collapse;

margin-top:20px;

}

table,th,td{

border:1px solid black;

padding:10px;

}
```

Upload.php

```
<?php

include("db.php");

libxml_use_internal_errors(true);

$xmlString = $_POST['xml'];

$dom = new DOMDocument();

if (!$dom->loadXML($xmlString)) {

    echo "<h2 style='color:red'>XML is NOT Well Formed</h2>";

    foreach(libxml_get_errors() as $error){
        echo $error->message."<br>";
    }

    exit();
}

echo "<h2 style='color:green'>XML is Well Formed</h2>";

if(!$dom->schemaValidate("student.xsd"))
{
    echo "<h2 style='color:red'>XML does NOT follow XSD</h2>";
}
```

```
        foreach(libxml_get_errors() as $error){
            echo $error->message."<br>";
        }

        exit();
    }

    echo "<h2 style='color:green'>XSD Validation Successful</h2>";

    $xml = simplexml_load_string($xmlString);

    $name = (string)$xml->Name;
    $roll = (int)$xml->RollNumber;

    $conn->query("INSERT INTO students(student_name,roll_number)
VALUES('$name','$roll')");

    $student_id = $conn->insert_id;

    foreach($xml->Subjects->Subject as $subject)
    {

        $subjectName = (string)$subject->Name;
        $marks = (int)$subject->Marks;

        $conn->query("INSERT INTO subject_marks(student_id,subject_name,marks)
VALUES($student_id,'$subjectName',$marks)");
    }

    echo "<h2>Student Uploaded Successfully</h2>";

    echo "<a href='index.php'>Back</a>";
?>
```